Closing Gaps: A Model of Cumulative Curation and Preservation Levels for Trustworthy Digital Repositories

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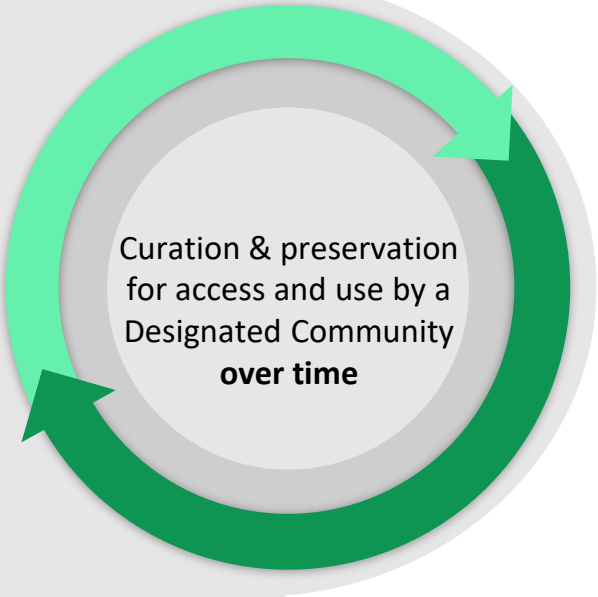
Curation and preservation

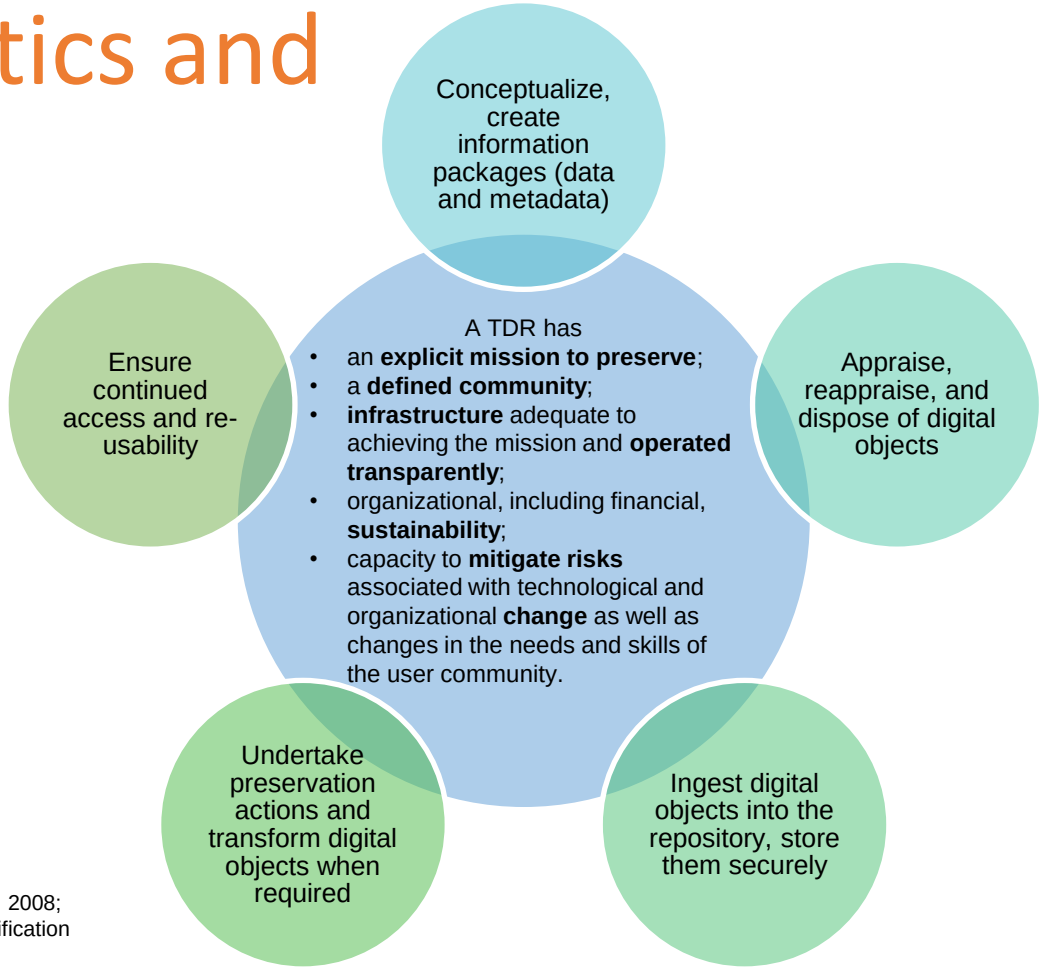
- Organizational framework (policies, resources, mission, etc.)
- Collection characteristics (e.g. volume, object types and formats, etc.)
- Designated Community needs, preferences, knowledge base



When it comes to "adequate" curation and preservation, one size does not fit all

→ depends on objects, community, objective, and more

TDR characteristics and lifecycle actions



CoreTrustSeal Standards and Certification Board, 2022; Higgins, 2008; International Organization for Standardization, 2012; nestor Certification Working Group, 2013; Research Libraries Group, 2002.

Why define curation levels?

- Community agreement on definition of “levels of care” provided by a repository provide an important reference point, e.g. for
  - planning and conception of services,
  - transparent communication with stakeholders,
  - benchmarking and evaluation.

CoreTrustSeal Levels of Curation 2017-2025

Level	Description
A.	Content distributed as deposited.
B.	Basic curation – e.g. brief checking, addition of basic metadata or documentation
C.	Enhanced curation – e.g. conversion to new formats during ingest, enhancement of documentation and metadata
D.	Data-level curation – as in C above, but with additional editing of deposited data [for accuracy]

From CoreTrustSeal Guidance

2017-2019, 2020-2022

- Curation understood as "adding value by enhancing content"
- Initial deposits unchanged, edits only on copies

2023-2025

- Curation as prerequisite for "assuring long-term accessibility and understandability of data as the needs of the Designated Community change"
- Initial deposits unchanged, edits only on copies
- Sufficient metadata for independent use by Designated Community
- Measures for active preservation in place

CoreTrustSeal Standards and Certification Board. (2019). CoreTrustSeal Trustworthy Data Repositories Requirements 2020–2022 (v02.00-2020-2022). Zenodo. https://doi.org/10.5281/zenodo.3638211
CoreTrustSeal Standards and Certification Board. (2022). CoreTrustSeal Requirements 2023-2025 (V01.00). Zenodo. https://doi.org/10.5281/zenodo.7051012

Revised curation & preservation levels

Z. Content distributed as deposited.	Data content and supporting metadata are distributed to users exactly as they are provided by depositors. Data content and supporting metadata are stored for a given time period, or indefinitely.
D. Deposit Compliance	Data content and supporting metadata deposited are checked at the point of deposit for compliance with defined criteria.
C. Initial Curation	In addition to Level D., if these criteria are not met the digital objects are curated by the repository to meet the defined criteria.
B. Logical-Technical Preservation	In addition to D. and/or C. the repository takes long-term responsibility for ensuring that the data and metadata can be rendered as required by the designated community.
A. Conceptual preservation for understanding and reuse	In addition to B., the repository takes long-term responsibility that the data content and metadata can be independently understood by the designated community.

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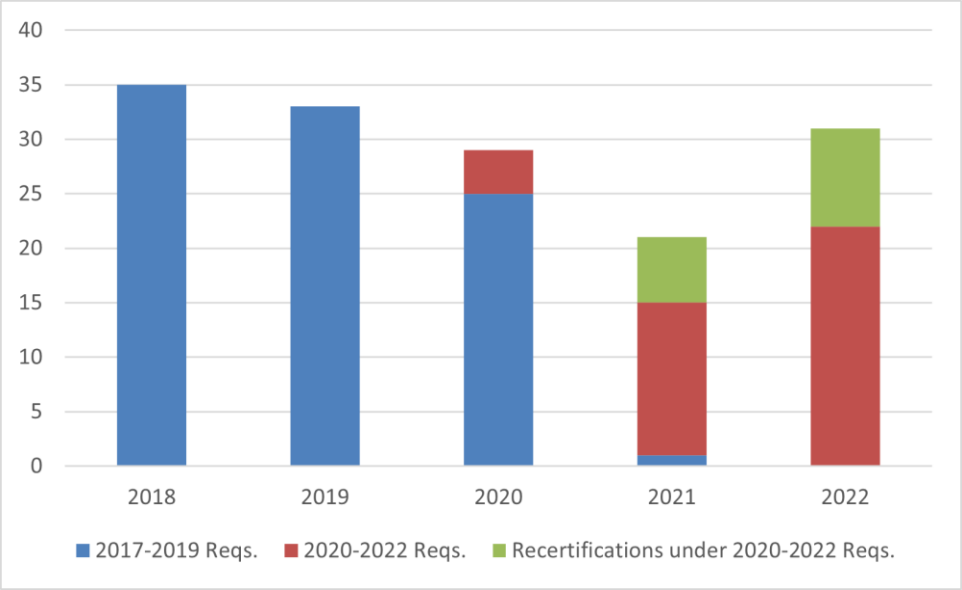
CoreTrustSeal Standards and Certification Board. (2023). Curation & Preservation Levels: CoreTrustSeal Discussion Paper (v02.00). Zenodo. https://doi.org/10.5281/zenodo.8083359

Analysis of 149 successful CoreTrustSeal applications 2018-2022

- Level(s) of curation offered by the applicant as stated under ‘R0. Level of Curation Performed’, including any explanations stated under ‘R0. Level of Curation Performed Comments’
- The response texts to ‘R8. Appraisal’ and ‘R10. Preservation Plan’ were analyzed to determine, among other things,
  - the presence/absence of ingest checks against a defined set of deposit criteria (based on R8. Appraisal),
  - the extent of initial curation carried out by the applicant at ingest of resources into the repository,
  - the preservation strategy employed (format migration, emulation, other/unspecified).

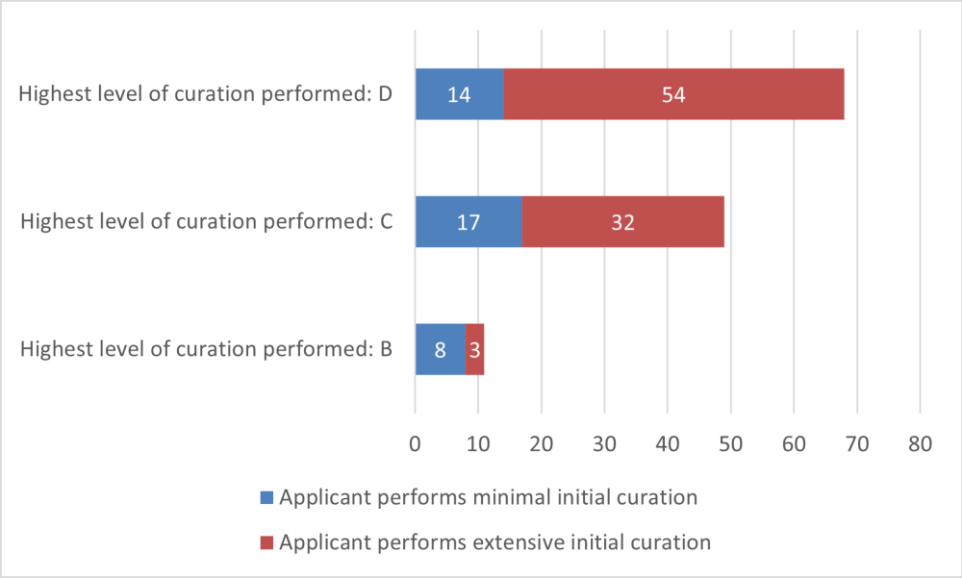
Sample



Number of certifications per year, differentiated by version of CoreTrustSeal Requirements (n = 149)

- 149 application
- 134 applicants
- 15 recertifications

Extent of initial curation



Highest level of curation and extent of initial curation (n=128)

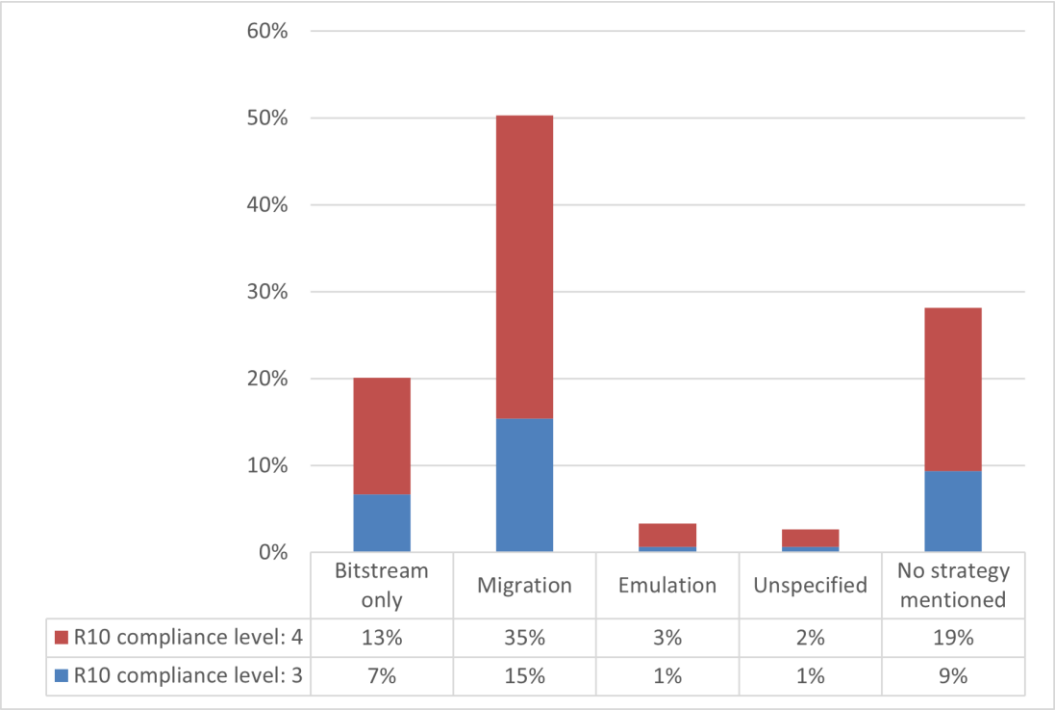
Minimal:

- correction of small errors in the (meta)data by repository staff,
- provision of support for data depositors in carrying out corrections of (meta)data before resubmission

Extensive:

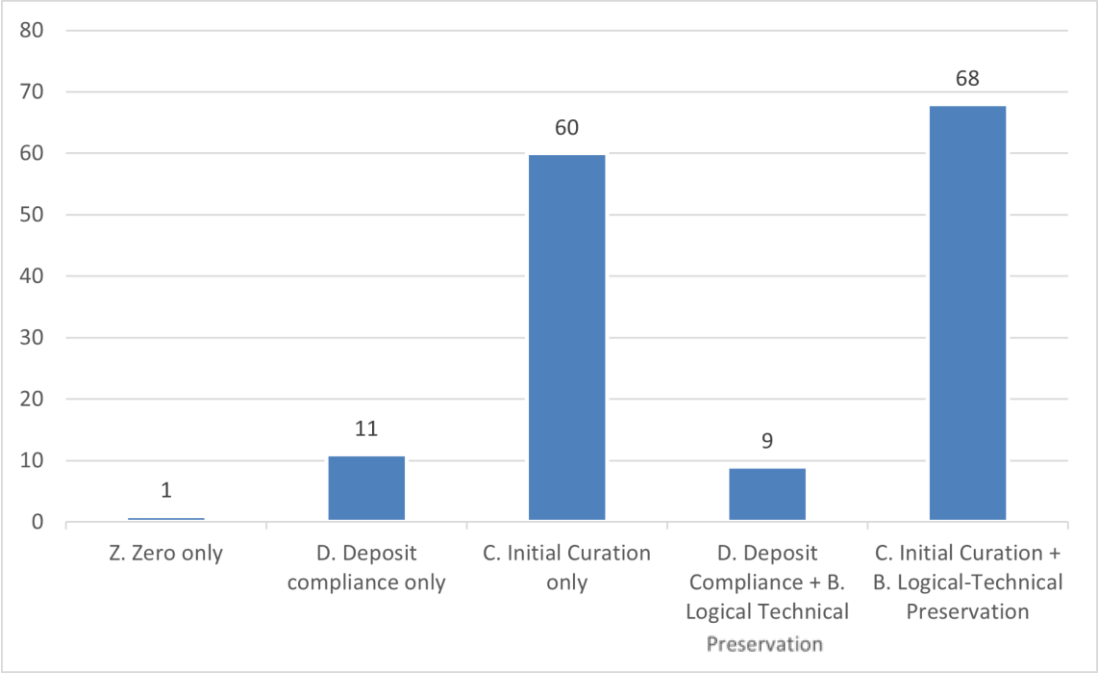
- creation of new metadata or documentation by repository staff,
- data harmonization and/or
- format conversions ('normalization') performed during ingest

Preservation strategies



Preservation strategies mentioned in R10 by % of applications (n=149)

Mapping on proposed levels



Maximum levels of curation and preservation achieved by applicants (n=149), derived by mapping to proposed model

Conclusion & next steps

- The proposed curation and preservation levels provide information about curation at ingest and preservation measures over time
  - more inclusive information about repository activities than current CoreTrustSeal Levels of Curation
  - reduce ambiguity during self-assessments, facilitate CoreTrustSeal in scope/out of scope decisions
- The levels have potential to be expanded to machine-actionable repository- and object-level metadata about services, curation, and preservation measures
- Next Steps:
  - Mapping current catalogues and lists of curation activities such as the DCN CURATE(D) Steps and Checklist (Data Curation Network, 2022) and the ARL SPEC Kit 354 (Hudson-Vitale et al., 2017) to proposed levels
  - Explore repository and object-level metadata implications from the curation and preservation levels

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